

SHIBANI PURBEY

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PROFILE

Results-driven Software Engineer with 4 years of experience designing and deploying high-performance web applications using Java, Spring Boot, Angular, and AWS, with strong expertise in CI/CD, cloud integration, and API development.

TECHNICAL SKILLS

- **Languages & Frameworks:** Java (8+), Spring Boot, Angular, JavaScript, Spring MVC, JPA, Hibernate, Maven
- **Architecture:** Microservices, RESTful Web Services, Design Patterns, SOLID Principles, Multi-tier Architecture
- **Databases:** PostgreSQL, MySQL, CouchDB, NoSQL, Oracle, JDBC
- **Cloud & DevOps:** AWS (ECS, EC2, S3, Lambda, RDS, API Gateway, ALB), Docker, Jenkins, Git, GitHub Actions
- **Security and AI:** OAuth 2.0, JWT, Spring Security, Generative AI, LLM Integration
- **Testing & Quality:** JUnit 5, Mockito, SonarQube, SAST & DAST Tools, Postman/Swagger
- **Messaging & Streaming:** RabbitMQ, Kafka, Asynchronous Programming, Event-Driven Architecture
- **Methodologies & Practices:** Agile/Scrum, Code Review, TDD, SLA Management

EDUCATION

Vellore Institute of Technology

B. Tech - Computer Science and Engineering

- CGPA: 8.64/10

Vellore, Tamil Nadu, India

Dec 2019 – July 2023

EXPERIENCE

Software Engineer II

Caterpillar Inc.

Aug 2026 – Present

Bangalore, India

- Developed the Quick Buy feature for Caterpillar's parts e-commerce platform (**parts.cat.com**), reports to process over **\$15M** in daily sales.
- Implemented **payment orchestration** for the e-commerce checkout flow, integrating payment gateways, transaction-status handling, retry logic, and refund workflows to coordinate payments with order placement.
- Owned **end-to-end feature enhancement**, gaining hands-on experience in software architecture, microservice integration, and cross-team collaboration.

Software Engineer

Baker Hughes

Oct 2024 – Aug 2026

Bangalore, India

- Served as **Technical Lead** for a team of **6** developers, driving day-to-day decisions and task prioritization, and delivered **3 production go-lives** through coordinated deployments and post-release stability.
- Developed and deployed Spring Boot microservices on AWS ECS, supporting an offline-first **PWA** used by **~8K** Field Service Engineers and strengthened codebase quality by resolving **100+** critical security issues.
- Enhanced application reliability by **80%** through strategic debugging, comprehensive testing frameworks (JUnit/Mockito), and robust **error handling implementation**.
- **Mentored interns** to deliver Angular and Java upgrades, avoiding **~\$38K** in **obsolescence** remediation costs.

Software Engineering Specialist

General Electric – GE Vernova

Jan 2023 – Sep 2024

Bangalore, India

- **Spearheaded** the development of a scheduling algorithm for gas turbine outages, delivering **\$30M** benefits.
- Built Spring Boot REST APIs with exception handling, DTO mapping, and data persistence using Spring Data JPA.
- Engineered an Angular–Spring Boot application with AWS monitoring, leveraging **Java concurrency** to improve application performance by **20%** while ensuring scalability and reliability.
- Achieved an impressive **85% code coverage** through unit testing, leading to substantial decrease in failure rates.

PROJECTS

- **Athena Onboard – AI-Powered HR Onboarding Automation**

AWS Lambda, API Gateway, Amazon Titan (Bedrock), S3, DynamoDB, SES, CloudWatch

- Built an AI-powered, serverless onboarding platform to auto-generate personalized training plans with HR review, audit traceability, and automated email delivery.
- Delivered 12x–24x projected ROI, reducing manual HR effort by ~20% and enabling \$4–\$8M annual cost savings through scalable automation.

- **E-commerce Application**

Spring Boot, Auth Server, MySQL, Eureka, Config Server, Zipkin, Kafka, Elasticsearch.

- Designed an online shopping platform with integrated product, order, notification and inventory services.
- Implemented synchronous and asynchronous communication, distributed tracing, centralized logging and authorization server.

- **Job Portal Web Application**

Spring Boot 3, Thymeleaf, MySQL, Spring Security

- Developed a full stack job portal enabling recruiters to post/manage jobs and candidates to search, apply, save jobs, upload resumes, and maintain profiles.
- Implemented role-based authentication, Spring MVC and Data JPA for backend logic & database persistence.

ACHIEVEMENTS

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| • Secured 1st place in the Baker Hughes + Capgemini Fueling Future Challenge, a GenAI Hackathon | 2025 |
| • Recipient of COMPEX Scholarship by Indian Embassy to study B. Tech CSE in VIT, Vellore | 2019 |
| • Merit Scholarship for Outstanding Performance in Academics, VIT Vellore | 2021 |
| • Registered Engineer (Computer Engineering), Nepal Engineering Council | 2024 |
| • Java and Spring Boot Certificate by Udemy link | Dec 2023 |
| • Associate Reactive Developer OutSystems link | Dec 2023 |

ENGAGEMENTS AND INTERESTS

- Volunteer, Karunashraya – a cancer hospice dedicated to providing compassionate care, Bangalore.
- Program Facilitator for PAHUNCH Project, Child Sensitive Social Protection (CSSP) by Save The Children.
- Enumerator, National Population & Housing Census 2021, Nepal.
- Volunteer of Janakpur Youth Network, Janakpur, Nepal